

PRICE MONITOR



CSE 100: Software Development Project I

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Abstract

The proposed software application is a government-initiated tool aimed at improving transparency and providing citizens with easy access to accurate product price information. This system addresses the limitations of manual price monitoring methods by offering a digital solution that enhances efficiency, reduces errors, and empowers users with timely data.

The application supports two distinct user roles: Admins and Users. Admins are responsible for managing product information, including adding, updating, and categorizing products, while Users can search for products, explore category-specific listings, and view detailed price histories. A key focus of the system is ensuring user accessibility through a simple and intuitive interface.

Developed in C++ and utilizing file handling for data storage, the application ensures compatibility with basic computer hardware, making it widely accessible without requiring advanced technology. Key features include secure user authentication, streamlined product management, comprehensive price tracking, and analytics for better decision-making.

This tool is designed to address the challenges of price manipulation, lack of transparency, and inefficiencies in traditional methods. By providing an accessible and user-friendly system, it empowers citizens to make informed decisions and promotes accountability in price regulation, creating a foundation for a more transparent and equitable market system.

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Chapter 1

INTRODUCTION

1.1. Problem Specification

i. Introduction

In today's digital age, having access to accurate and timely product pricing information is crucial for ensuring transparency and fairness in market practices. Citizens rely on such data to make informed decisions, while governments depend on efficient systems to regulate markets and maintain trust. The "Government Product Price Management System" aims to address these needs by creating a user-friendly, centralized platform that bridges the gap between government-managed pricing data and public accessibility. This system is designed to simplify the dissemination and management of product prices, fostering transparency and accountability.

ii. Problem Statement

Governments often face challenges in managing and communicating product prices due to outdated, manual processes and the lack of centralized digital systems. These inefficiencies make it difficult for citizens to verify price authenticity, creating opportunities for misinformation and market manipulation. At the same time, administrators find it cumbersome to maintain accurate, up-to-date records and secure data against errors or tampering. The absence of a streamlined, accessible system undermines trust between governments and citizens and hampers effective market regulation. A robust, automated solution is needed to address these shortcomings by ensuring accuracy, accessibility, and real-time updates.

iii. Motivation

The motivation behind this project is rooted in the need to modernize the traditional approaches to price management and information sharing. In developing countries, inconsistent market regulation and limited access to reliable pricing information often leave consumers vulnerable to exploitation. By introducing a centralized, automated system, this project seeks to empower citizens with accurate, accessible information while enabling administrators to efficiently manage and update price records. This initiative supports the broader vision of digital governance, reducing inefficiencies and promoting fairness and trust in the market.

iv. Significance

The "Government Product Price Management System" represents a transformative step toward improving governance and public service delivery. By ensuring transparency in pricing information, the system can build trust between governments and citizens while fostering market integrity. Its implementation embodies the principles of good governance, such as accountability, efficiency, and inclusivity. Furthermore, the system's scalable design makes it adaptable to diverse regions and

contexts, providing a sustainable model for digital transformation in public administration.

1.2 Objectives

- To develop a secure and accessible system for managing product prices.
- To enable government administrators to add, update, and monitor product prices efficiently.
- To provide citizens with accurate and real-time access to product prices.
- To facilitate category-wise product listings and detailed price analytics.
- To promote transparency and accountability in governmental price management.
- To create a scalable and user-friendly application.

1.3 Organization of Project Report

This report is structured into several chapters, each addressing key aspects of the project:

- **Chapter 1: Introduction** – Introduces the problem, motivation, significance, objectives, and organization of the report.
- **Chapter 2: Background** – Explores existing systems and methodologies related to price management.
- **Chapter 3: System Analysis & Design** – Describes the architecture, design principles, and implementation details of the proposed system.
- **Chapter 4: Implementation** – Provides a detailed account of the development process, including code structure and functionality.
- **Chapter 5: User Manual** – Describes and illustrates the User Interfaces of both Admin and User and provides sample credentials to enter into the system.
- **Chapter 6: Conclusion and Future Work** – Summarizes the project's achievements and proposes future enhancements.

Each chapter is designed to provide a comprehensive understanding of the project, from its conceptualization to implementation and evaluation, ensuring clarity and coherence throughout the report.

Chapter 2

BACKGROUND

2.1. Existing System Analysis

In today's world, effective solutions are necessary to simplify and enhance everyday activities such as monitoring and controlling pricing information. There are no such digital systems available in our country to monitor product prices. Our government uses a manual system which limits these critical pieces of information from accessing by the citizens.

Manual Systems

Manual systems typically involve physical records, such as spreadsheets, written records, or notice boards, where authorities list product prices.

Pros:

1. **Simplicity:** Easy to set up without requiring complex technology.
2. **Accessibility:** Ideal for areas with minimal technical literacy or no access to the internet.
3. **Cost-effective:** No upfront investment in software or hardware.

Cons:

1. **Inefficiency:** Updating product price records manually requires significant time and effort, making the process slower and more prone to mistakes caused by human error.
2. **Lack of Accessibility:** Citizens can not easily access these records due to distance or other security procedures.
3. **Data Security:** Physical records are vulnerable to tampering, accidental loss, or damage due to mishandling, environmental factors, or unforeseen incidents, leading to potential inaccuracies or loss of critical information.

2.2. Research Gap

Despite the clear advantages of a digital approach, there is no existing system in our country that offers real-time price monitoring and analytics for citizens. The current manual system used by the government is limited in accessibility, efficiency, and scalability.

The gaps in the existing system include:

1. **Absence of Digital Solutions:** No digital platform currently exists that consolidates product pricing information for public access. Citizens are dependent on manual

methods like news, which are time-consuming and often outdated.

2. **Inefficiency in Updates:** Manual systems are slow and error-prone when updating or disseminating pricing information. This creates delays and inconsistencies, particularly in urgent scenarios.
3. **Limited Accessibility:** Physical notice boards or manual records are not readily accessible to all citizens, especially those in remote areas or with mobility constraints.
4. **Transparency and Accuracy:** Manual records lack sufficient checks to ensure accuracy and integrity, leading to potential misinformation or manipulation.
5. **Scalability and Usability:** The manual system is challenging to scale for broader use, and any attempts to digitize processes lack user-friendly features that cater to a diverse audience.

This project aims to address these gaps by creating a streamlined digital system that allows real-time updates, robust data analytics, and a user-friendly interface. It seeks to enhance accessibility, transparency, and efficiency while empowering citizens with critical pricing information. This solution will serve as a foundation for bridging the gap between manual inefficiencies and modern digital capabilities.

Chapter 3

SYSTEM ANALYSIS & DESIGN

3.1. Technology & Tools

This console-based system was developed using a combination of technologies and tools, focusing on simplicity, efficiency, and maintainability. The following details the software and hardware requirements, tools, and technologies employed during development.

Software Requirements

1. **Programming Language:**
 - **C++:** Chosen for its robustness, efficiency, and extensive support for object-oriented programming.
2. **Development Environment:**
 - **IDE:** Code::Blocks 20.03, configured with GCC Compiler 15.0.0 for compiling and running the C++ code.
3. **Version Control:**
 - **Git:** To track changes and collaborate effectively throughout the development cycle.
4. **Operating System:**
 - Windows 10/11 or Linux distributions that support GCC and Code::Blocks.

Hardware Requirements

1. **Processor:** Dual-core processor or higher (Intel Core i3 or equivalent).
2. **RAM:** Minimum 4GB (8GB recommended for smooth operation).
3. **Storage:** At least 500 MB of free space for the IDE, compiler, and project files.
4. **Display:** A monitor with a resolution of 1366x768 or higher.

Tools and Technologies

1. **File Handling in C++:**
 - Used extensively to store, retrieve, and update data, ensuring the persistence and reliability of records.
2. **Error Handling:**
 - Integrated basic error handling mechanisms to manage file read/write operations and invalid user inputs.
3. **Text Editors:**
 - **Notepad++:** This is for quick code reviews and modifications outside the IDE.

Additional Considerations

The selected tools and technologies were aimed at achieving a balance between functionality and simplicity. The reliance on widely available tools ensures that the system can be easily replicated or modified for future needs.

3.2. Diagram & Flowchart

Below is a simplified flowchart that demonstrates the operation of the system

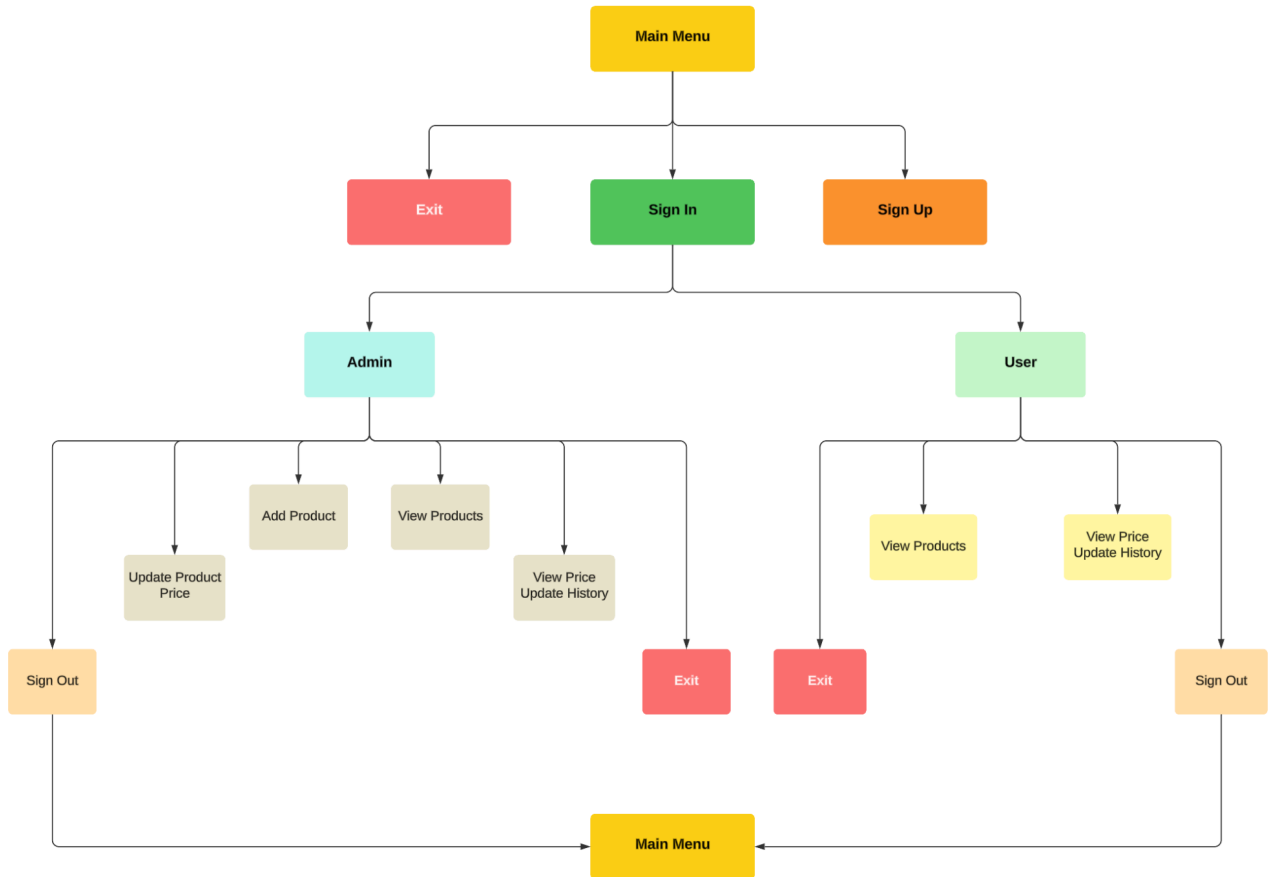


Fig 1: Project Flow Diagram

Chapter 4

IMPLEMENTATION

4.1. Interface/Front-End Design

The interface of this project is entirely console-based, designed to ensure simplicity and clarity for both administrators and users. Using standard input and output functions in C++, menus were implemented to guide users through various operations. The system relies on a text-based interface, with prompts and options displayed for user interaction. Each menu and sub-menu follows a structured format, ensuring users can navigate easily. Special care was taken to ensure input validation, providing error messages when invalid data is entered.

4.2 Functionalities/Back-End Design

The back-end design of this project is implemented in C++, using file handling to store and manage data persistently. Text files were used as the database to save user credentials, product details, and price update logs. The core functionalities include reading from and writing to these files, ensuring data consistency.

- **User Authentication:** The system allows users to register and log in, differentiating between Admin and regular User roles. User data, including email and password, is stored in a file to maintain security.
- **Product Management:** Admins can add new products, and update their prices. Each action involves updating the product file while maintaining a history of price changes for analytics.
- **Data Persistence:** Files are used to store information persistently, ensuring that the data remains available even after the program exits. Functions were developed to handle reading, writing, appending, and searching within these files efficiently.
- **Analytics:** Logs are maintained to track all changes made to product prices, providing users and admins with insights into trends and updates..

4.3 Modules/Features

The project is broken up into multiple modules, each of which carries out distinct functions. Among the primary modules are:

1. **Authentication Module:**
 - Handles user registration and login.
 - Differentiates between Admin and User roles.
 - Stores user credentials securely in a file.
2. **Product Management Module:**
 - Admin can add and update product details.
 - Maintains a log of all price changes.

- Stores product data, including name, category, and price, in a text file.
- 3. **File Handling Module:**
 - Implements functions to read, write, and update files.
 - Ensures data consistency and handles concurrency issues.
- 4. **Analytics Module:**
 - Logs product updates and displays trends in price changes.
 - Enables users to analyze price variations over time.
- 5. **User Interface Module:**
 - Provides a text-based menu system for navigation.
 - Ensures ease of use with clear prompts and messages.

The implementation emphasizes modularity and scalability, making it easier to debug and expand in the future. Each module interacts seamlessly with others, providing a cohesive user experience while ensuring the program remains efficient and reliable.

Chapter 5

USER MANUAL

5.1. System Requirements

5.1.1. Hardware Requirements

To successfully set up and run the console-based project, the following hardware configuration is recommended:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** At least 500 MB of free space
- **Display:** 1366x768 resolution or higher
- **Input Devices:** Keyboard and mouse

5.1.2. Software Requirements

The following software configuration is necessary to set up and run the project:

- **Operating System:** Windows 10 or higher / Linux (Ubuntu 20.04 or higher)
- **Development Environment:** GCC compiler (Version 15.0.0 or higher)
- **Libraries/Dependencies:** Standard C++ libraries
- **File Handling Setup:** Ensure read/write access to the working directory for file storage.

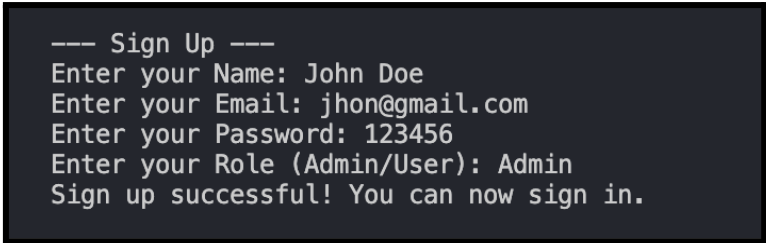
5.2. User Interfaces

This section provides an overview of the project's modules with screenshots and user interface descriptions. The project consists of two main panels: the **Admin Panel** and **User Panel**.

5.2.1. Admin Panel

The Admin Panel allows the admin to perform the following actions:

- **Sign In/Sign Up:**
Admins can log in using their registered credentials or sign up for a new account.



```
--- Sign Up ---  
Enter your Name: John Doe  
Enter your Email: jhon@gmail.com  
Enter your Password: 123456  
Enter your Role (Admin/User): Admin  
Sign up successful! You can now sign in.
```

Fig 2: Sign Up for Admin

```
--- Sign In ---  
Enter your Email: jhon@gmail.com  
Enter your Password: 123456
```

Fig 3: Sign In for Admin

- **Admin Menu:**

This is the interface Admins will see after signing in to the system. From here, admins can operate all the available admin functionalities.

```
--- Admin Menu ---  
1. Add Product  
2. View Products  
3. Update Product Price  
4. View Price Update History  
5. Sign Out  
6. Exit System  
Enter your choice: 2
```

Fig 4: Admin Menu

- **Add New Products:**

Admins can add new products to the list, specifying their names, categories, and prices.

```
--- Add Product ---  
Enter Product Name: Mayo  
Enter Product Price: 100  
Enter Product Category: Vegetables  
Product added successfully with ID: 32!
```

Fig 5: Add Product

- **Update Product Prices:**

Admins can modify the prices of existing products.

```
--- Update Product Price ---  
Enter Product ID: 32  
Current Price of Mayo is 100  
Enter New Price: 120  
Product price updated successfully!
```

Fig 6: Update Product Price

- **View Logs:**

Admins can view logs to track price changes and user activities.

```

--- Price Update History ---
Enter Product ID: 1

Price Update Log for Product ID: 1

```

Date	Old Price	New Price	Change (%)
Mon Dec 9 20:50:40 2024	120	75	-37.5%
Mon Dec 9 12:44:29 2024	75	120	+60.00%
Mon Nov 11 23:30:42 2024	80	75	-6.25%
Mon Nov 11 22:57:08 2024	70	80	+14.28%
Mon Nov 11 22:56:16 2024	60	70	+16.66%
Overall Rate:			+30.32%

Fig 7: Price Update History for Admin

5.2.2. User Panel

The User Panel provides access to the following functionalities:

- **Sign In/Sign Up:**
Users can create accounts or log in to their existing accounts.

```

--- Sign In ---
Enter your Email: abdullahsakib888@gmail.com
Enter your Password: 123456

```

Fig 8: Sign-In for Users

```

--- Sign Up ---
Enter your Name: Abdullah Sakib
Enter your Email: abdullahsakib888@gmail.com
Enter your Password: 123456
Enter your Role (Admin/User): User
Sign up successful! You can now sign in.

```

Fig 9: Sign Up for Users

- **User Menu:**
This is the interface the user will see after signing in to the system. Users will be able to operate all the available user features from here.

```

--- User Menu ---
1. View Products
2. View Price Update History
3. Sign Out
4. Exit System
Enter your choice: 1

```

Fig 10: User Menu

- **View All Products:**
Users can view all products, their prices, and categories.

```

--- Product List ---
ID   Name           Price    Category
-----
1    Potato          75.00   Groceries
2    Eggs             14.00   Groceries
3    Onion            120.00  Groceries
4    Milk             50.00   Dairy
5    Bread            30.00   Bakery
6    Apples           90.00   Fruits
7    Chicken          200.00  Meat
8    Rice             70.00   Groceries
9    Butter           80.00   Dairy
10   Tomatoes         40.00   Vegetables
11   Cheese           120.00  Dairy
12   Carrots          35.00   Vegetables
13   Bananas          60.00   Fruits
14   Beef             300.00  Meat
15   Flour            40.00   Bakery
16   Sugar            45.00   Groceries
17   Orange Juice     150.00  Beverages
18   Strawberries     180.00  Fruits
19   Peas             50.00   Vegetables
20   Pasta            75.00   Groceries
21   Fish             250.00  Meat
22   Cabbage          30.00   Vegetables
23   Yogurt           60.00   Dairy
24   Chocolate        90.00   Snacks
25   Coffee           250.00  Beverages
26   Lettuce          25.00   Vegetables
27   Mangoes          120.00  Fruits
28   Ice Cream        200.00  Dairy
29   Tea              180.00  Beverages
30   Biscuits         70.00   Snacks
31   Mayo             100.00  Vegetables

```

Fig 11: Products List

- **View Price Update History:**
Users can see the history of each product.

```

--- Price Update History ---
Enter Product ID: 1

Price Update Log for Product ID: 1
Date           Old Price    New Price    Change (%)
-----
Mon Dec 9 20:50:40 2024    120         75          -37.5%
Mon Dec 9 12:44:29 2024     75         120         +60.00%
Mon Nov 11 23:30:42 2024     80          75          -6.25%
Mon Nov 11 22:57:08 2024     70          80          +14.28%
Mon Nov 11 22:56:16 2024     60          70          +16.66%

Overall Rate:                                     +30.32%

```

Fig12: Price Update History for Users

5.2.3. Login Credentials

Admin Login Credentials:

- Username: admin@gmail.com
- Password: 123456

User Login Credentials:

- Username: user@gmail.com
- Password: 123456

Chapter 6

CONCLUSION

6.1. Conclusion

This project was created to close the gap between citizens' access to and the government's provision of goods price information. We were able to successfully develop an intuitive platform through the system that makes it easier for administrators and users to communicate. We obtained practical expertise with fundamental C++ concepts, such as file management and error handling, during the development process. Additionally, this project offered insights into modular programming, user interface design, and software development processes.

By providing a simplified and easily accessible solution, the project has shown to be successful in resolving the price transparency issue. All things considered, the process of creating this product has been instructive and fulfilling.

6.2. Limitations

Despite its success, the project has some limitations that need to be addressed:

1. **Scalability Issues:** The file-handling system may face challenges when dealing with large datasets, as it lacks advanced database management.
2. **Security Concerns:** The authentication system is basic, with minimal encryption, leaving it vulnerable to potential breaches.
3. **Limited User Interface:** Being a console-based system, the user interface lacks visual appeal and may not be as intuitive as modern graphical interfaces.
4. **No Real-Time Updates:** The system cannot efficiently handle real-time price updates or simultaneous user interactions.

6.3. Future Works

Several enhancements can be made to improve the system in the future:

1. **Database Integration:** Replace file handling with a database management system (e.g., MySQL or MongoDB) to improve scalability and performance.
2. **Graphical User Interface (GUI):** Develop a GUI using a framework like Qt or Tkinter to enhance usability and user experience.
3. **Enhanced Security:** Implement robust encryption and multi-factor authentication to secure user data and credentials.
4. **Real-Time Functionality:** Add support for real-time data updates and simultaneous multi-user access.
5. **Analytics Dashboard:** Introduce an analytics dashboard for admins to monitor trends, generate reports, and make informed decisions.
6. **Cross-Platform Compatibility:** Extend the system to run on multiple platforms, including mobile devices.

References

1. *Price Monitor Flowchart : Lucidspark.* (n.d.).
https://lucid.app/lucidspark/e68ebf62-1c40-4c17-95e5-5e9f4398efde/edit?viewport_loc=-2087%2C-719%2C4709%2C2441%2C0_0&invitationId=inv_20192406-461d-4139-b900-869358e84934

Appendix A